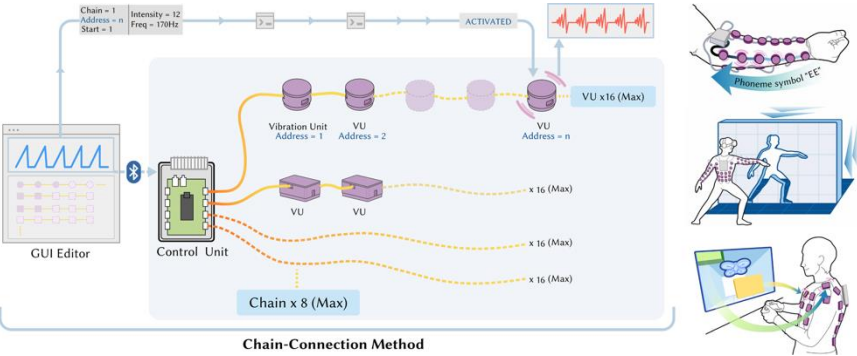


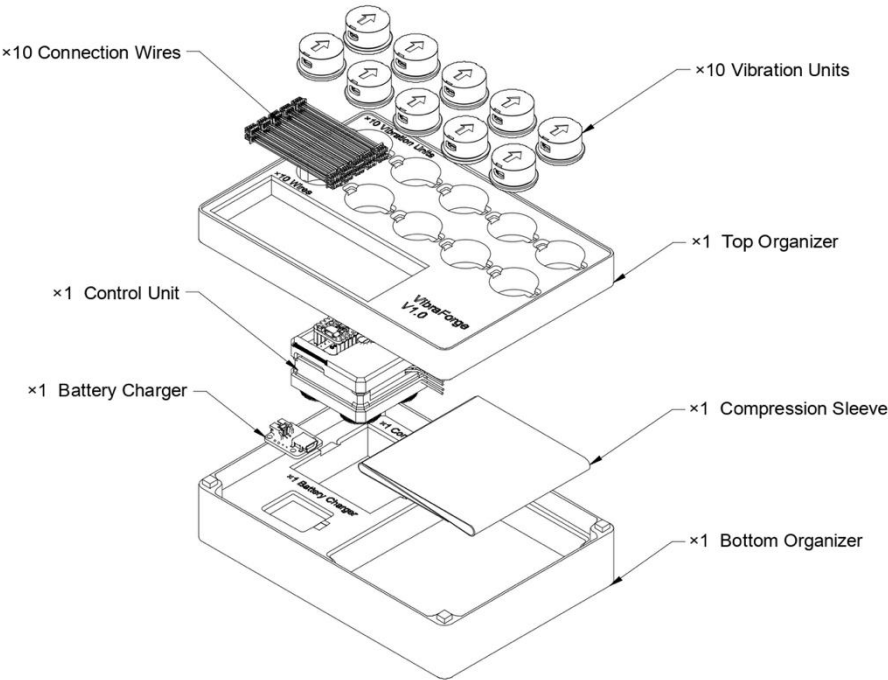
VibraForge Dev Kit v1

A Scalable Prototyping Toolkit For Creating Spatialized Vibrotactile Feedback Systems



Designed and Assembled by Qilong (Jerry) Cheng, April 2025

What's Inside?



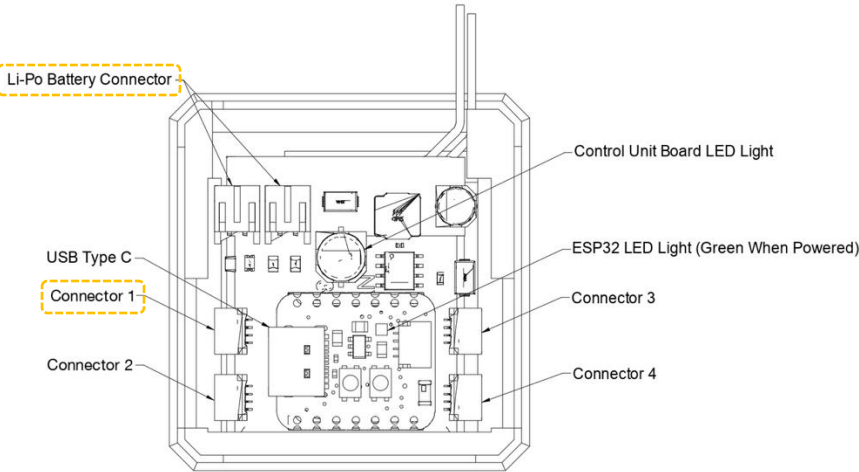
We're excited to help you begin your journey into the world of haptics!

Quick Start Guide:

Step 1 Control Unit

Connect the Li-Po battery to the **Li-Po Battery Connector** located at the top. A green LED will light up to indicate a successful connection.

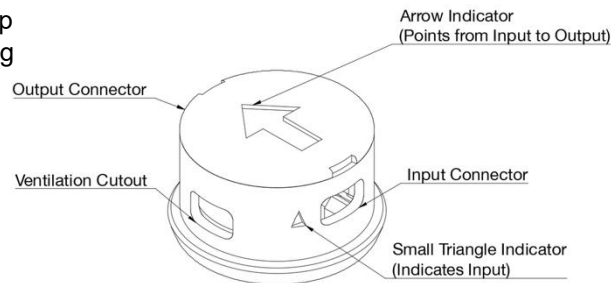
The diagram below shows four connectors for each vibration unit chain. We recommend starting with **Connector 1** for optimal configuration.



Step 2 Vibration Units

The arrow → on the cap indicates the connecting direction from input to output.

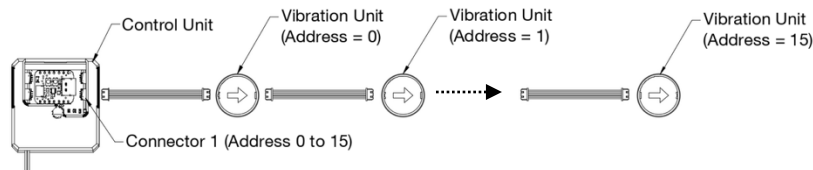
Do not reverse the direction, or the unit driver board might be damaged due to short circuit.



Step 3 Connect Them Together

Each chain supports up to 16 units. The unit address on each chain starts from 0.

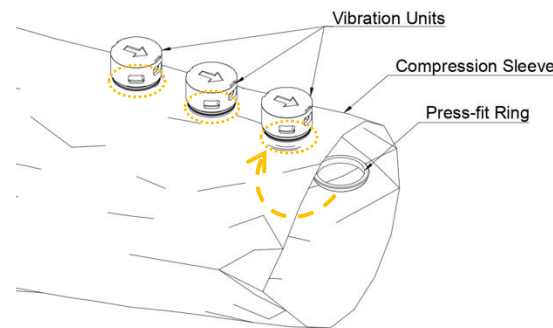
For example, the units on the first chain have address 0,1,2,...,15.



Step 4 Attach Vibration Units to the Sleeve

Slide the vibration unit **onto** the compression sleeve and place the press-fit ring **under**.

Press the components together until they securely lock onto the sleeve.



Step 5 Have Fun!

Once the hardware is setup, you can play vibrations using software from the GitHub Repo <https://github.com/huangbj16/VibraForge>

An easy place to start is to run Software_Design/Python_Server/Python_Test.py

- `pip install bleak`
- `python Python_Test.py`

If you prefer using GUI, try GUI_Editor/main app/app.py
- `pip install PyQt6 numpy matplotlib scipy bleak`
- `python app.py`



Useful Links:

A full walkthrough video for the assembly:

<https://www.youtube.com/watch?v=Ji4QbBA8m8U>

Open-source GitHub repository:

<https://github.com/huangbj16/VibraForge>

If you have any other questions for the toolkit or potential collaborations, feel free to email me!

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